AS · Cambridge (CIE) · Physics

15 mins 15 questions

Multiple Choice Questions

Turning Effects of Forces

Centre of Gravity / Moments / Turning Effects of Forces

Easy (5 questions)	/5
Medium (5 questions)	/5
Hard (5 questions)	/5
Total Marks	**/15**

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Easy Questions

1 What is the centre of gravity of an object?

A. the geometrical centre of the object

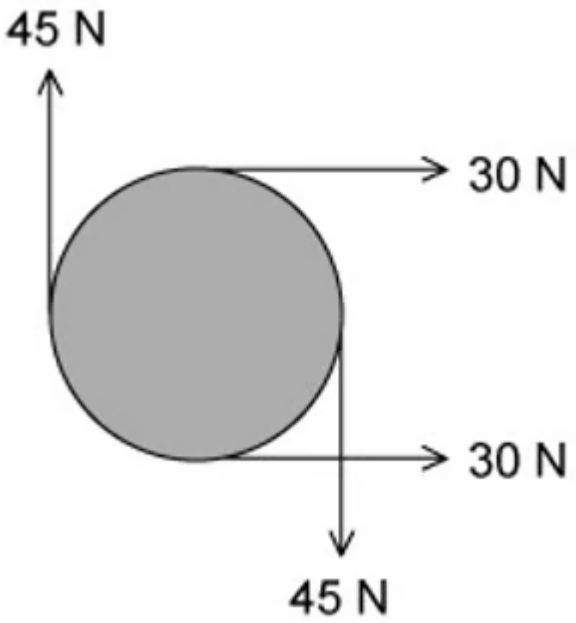
B. the point about which the total torque is zero

C. the point at which the weight of the object may be considered to act

D. the point through which gravity acts

(1 mark)

2 The diagram shows four forces applied to a circular object.



Which of the following describes the resultant force and resultant torque on the object?

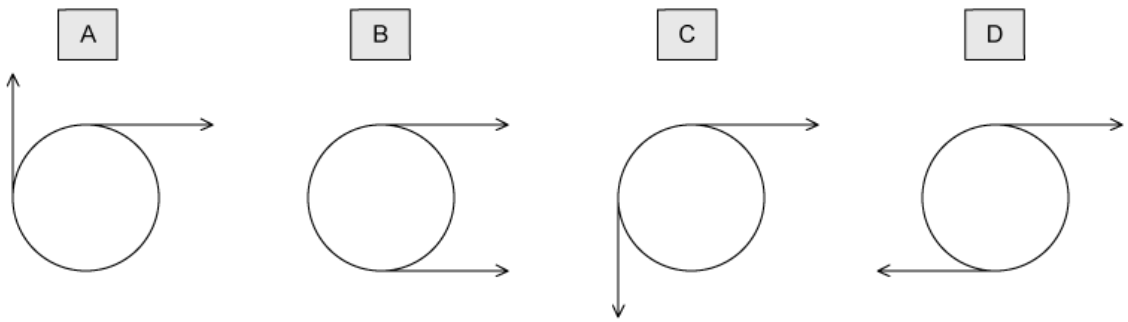
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	Resultant force	Resultant torque
A	non-zero	non-zero
B	non-zero	zero
C	zero	non-zero
D	zero	zero

(1 mark)

3 Two co-planar forces act on the rim of a wheel. The forces are equal in magnitude.

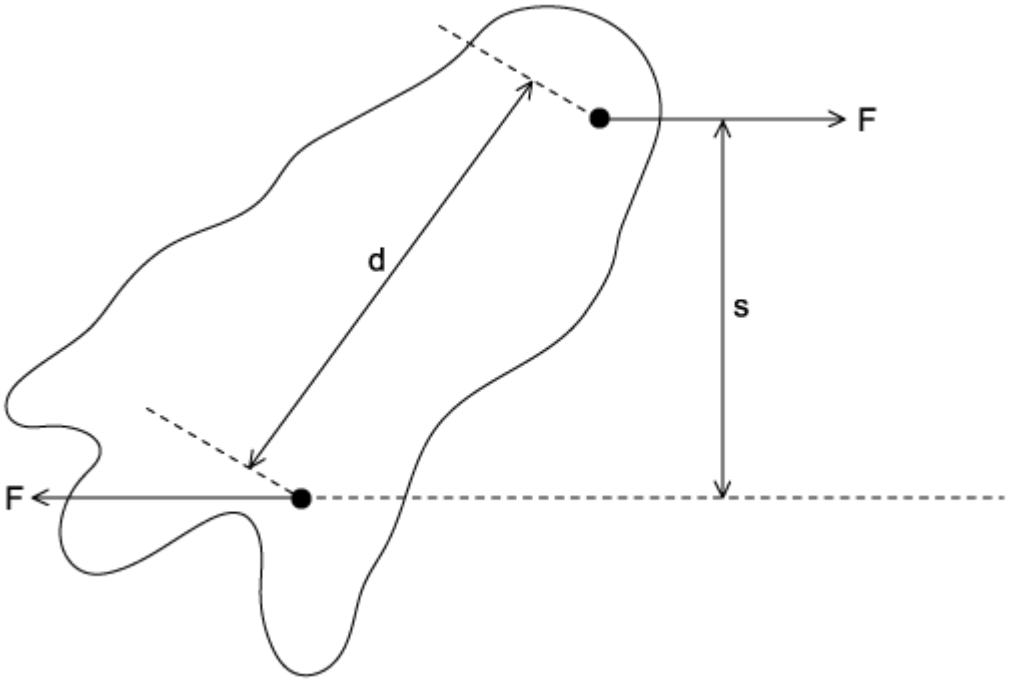
Which arrangement of forces provides only a couple?



(1 mark)

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4 Two parallel forces, each of magnitude *F*, act on a body as shown



What is the magnitude of the torque on the body produced by this couple?

A. *Fd*

B. *Fs*

C. 2 *Fd*

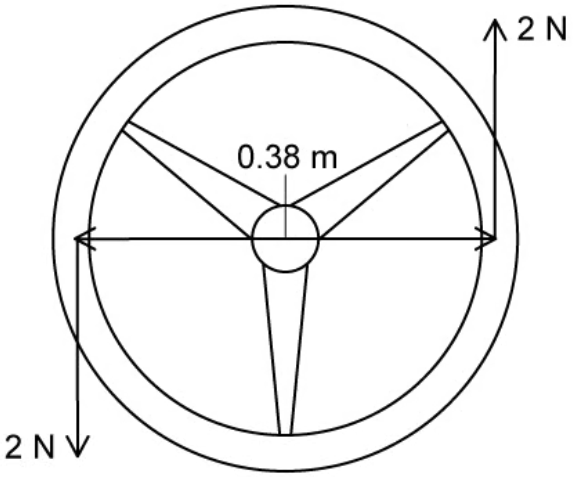
D. 2*Fs*

(1 mark)

5 A steering wheel of diameter 0.38 m is rotated to the left by two forces of 2 N, as shown in

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the diagram.



What is the torque of the couple produced by the rotation of the steering wheel?

A. 0.38 N m

B. 0.50 N m

C. 0.76 N m

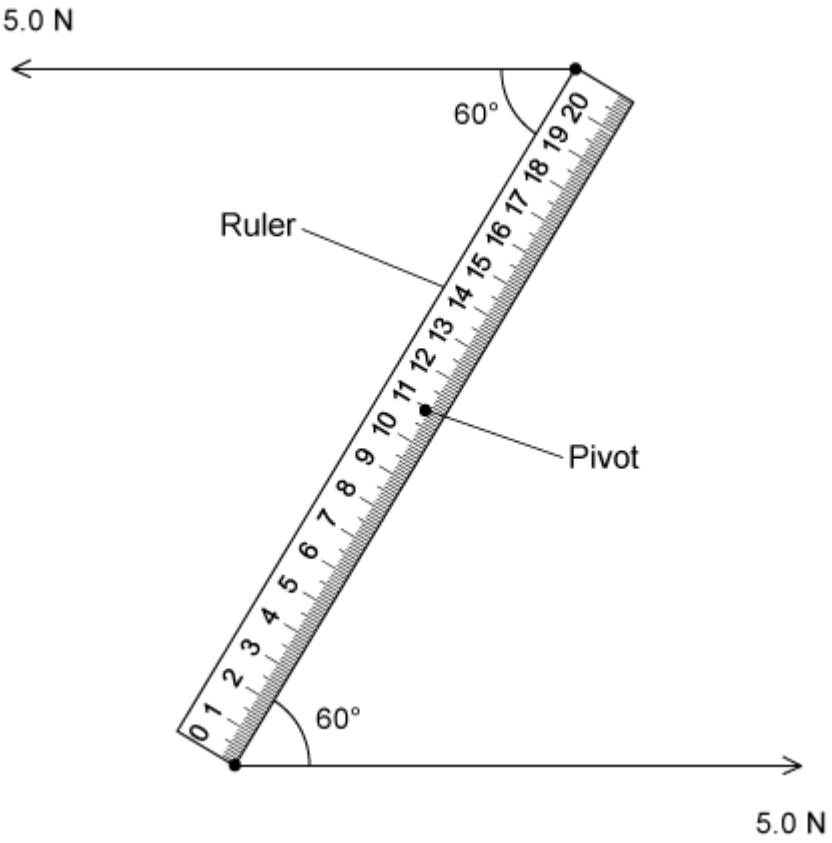
D. 1.0 N m

(1 mark)

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Medium Questions

1 A 20 cm ruler is pivoted at its centre. Equal and opposite forces of magnitude 5.0 N are applied to the ends of the ruler, creating a couple as shown



What is the magnitude of the torque of the couple on the ruler when it is in the position shown?

A. 0.2 N m

B. 0.9 N m

C. 1.0 N m

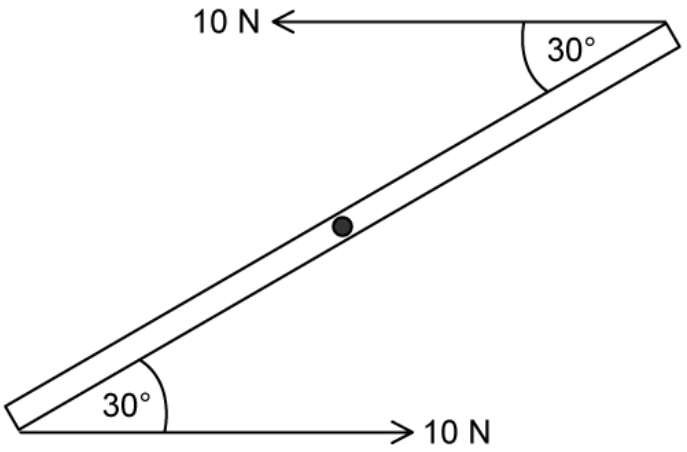
D. 1.7 N m

(1 mark)

2 A 10 cm beam is pivoted at its centre. Equal and opposite forces of 10 N pull the ends of the

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beam, creating a couple as shown.



What magnitude is the torque of the couple?

A. 0.3 N m

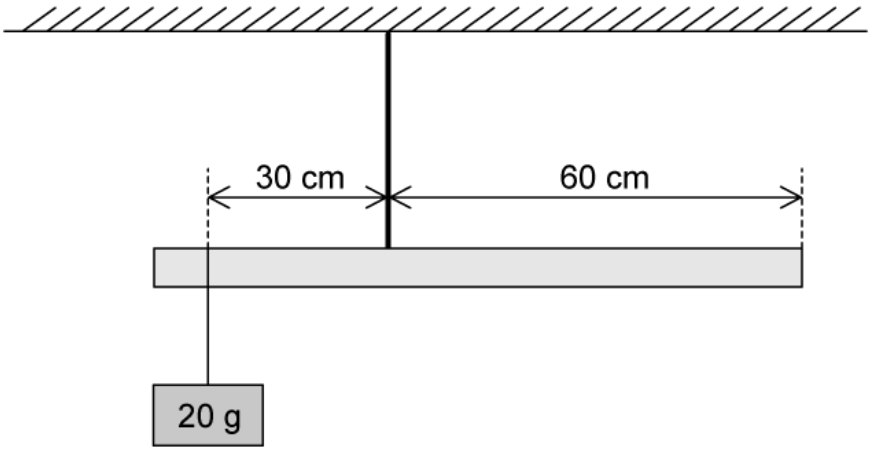
B. 0.5 N m

C. 0.9 N m

D. 1.0 N m

(1 mark)

3 A 1 m long uniform beam is hung from the ceiling by a rope. and is balanced when a 20 g mass is positioned 30 cm from the rope as shown.



What is the mass of the metre rule?

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A. 0.06 g

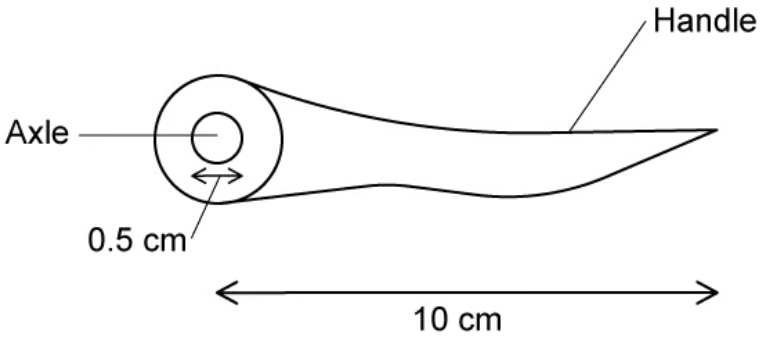
B. 20 g

C. 60 g

D. 100 g

(1 mark)

4 A door handle in the diagram below has a length 10 cm requires a force of 20 N to turn it.



If the handle breaks and the door must be opened using the axle of the handle, with a diameter of 0.5 cm, how much force is needed?

A. 4 N

B. 8 N

C. 40 N

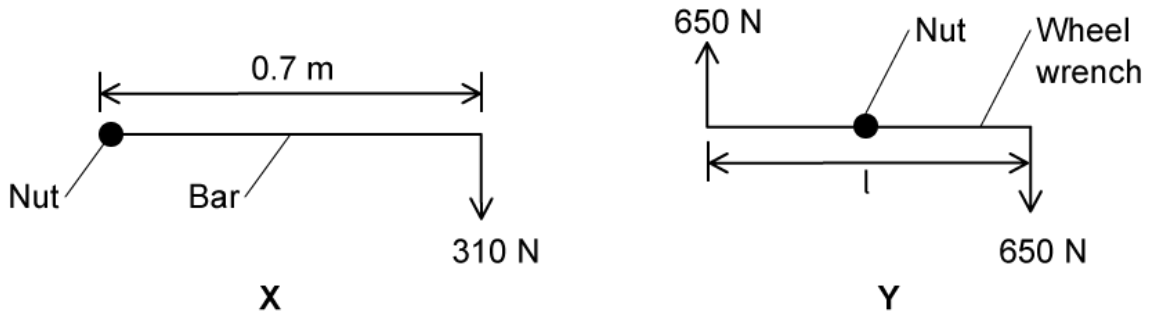
D. 400 N

(1 mark)

5 A car wheel nut can be loosened by applying a force of 310 N to the end of a bar of length 0.7 m, as in **X**. A car mechanic is capable of applying forces of 650 N simultaneously in

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opposite directions on the ends of a wheel wrench, as in **Y**.



What is the minimum length l of the wrench needed for the mechanic to loosen the nut?

A. 0.2 m

B. 0.3 m

C. 3.0 m

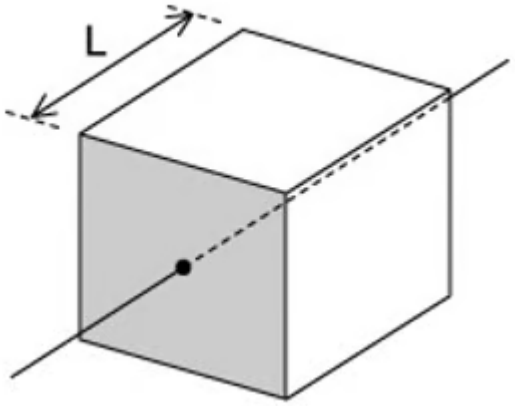
D. 6.0 m

(1 mark)

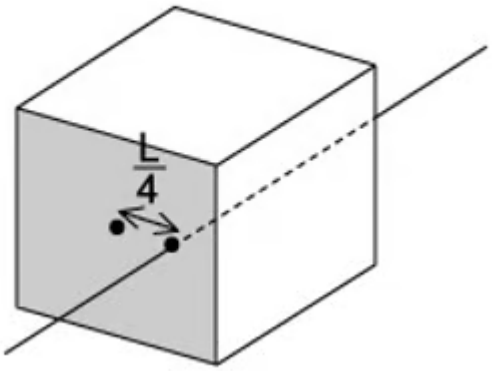
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Hard Questions

1 The diagram shows a solid cube with weight *W* and sides of length *L*. It is supported by a frictionless spindle that passes through the centres of two opposite vertical faces. One of these faces is shaded.



The spindle is now removed and replaced at a distance $\frac{L}{4}$ to the right of its original position.



When viewing the shaded face, what is the magnitude of the torque required to stop the cube from toppling?

A. $\frac{WL}{2}$ anticlockwise

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B. $\frac{WL}{2}$ clockwise

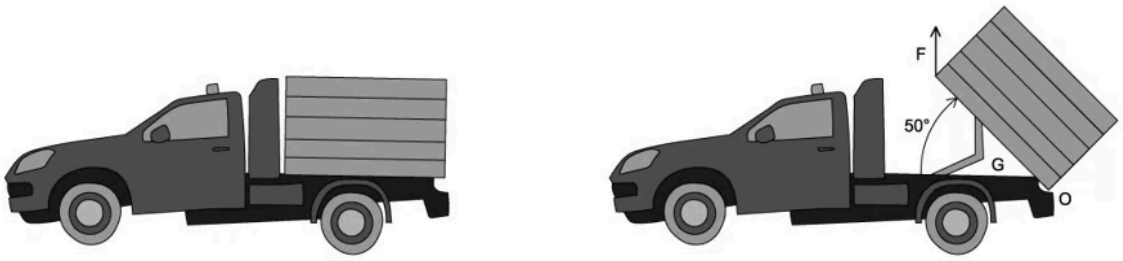
C. $\frac{WL}{4}$ anticlockwise

D. $\frac{WL}{4}$ clockwise

(1 mark)

2 A trailer of weight 30 kN is attached to a truck. The trailer is raised until it is held at 50˚ to the horizontal by a vertical force *F* as shown in the diagram.

The trailer is 2.7 m in length, and the position of its centre of gravity G is a third of the length from the hinge at O.



What is the vertical force *F* applied to the end of the trailer?

A. 10 kN

B. 15 kN

C. 30 kN

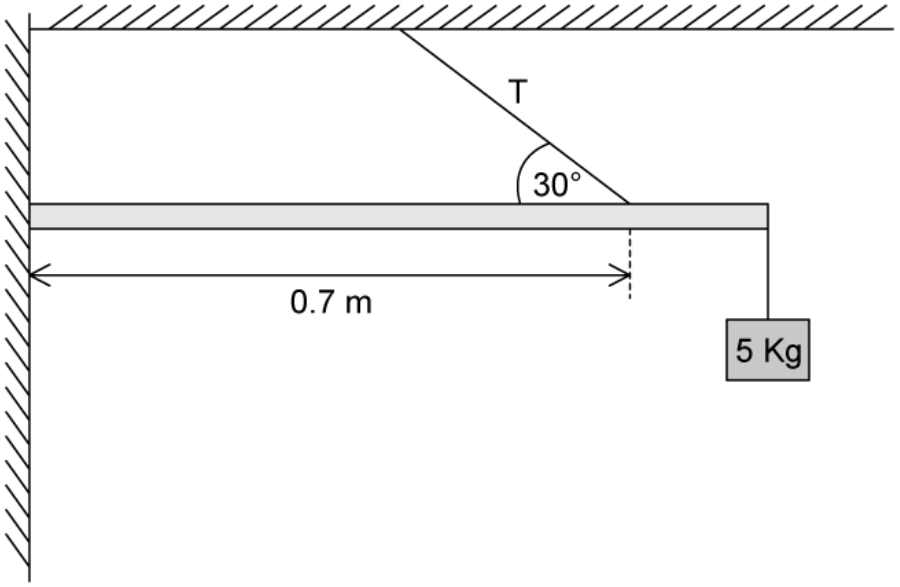
D. 60 kN

(1 mark)

3 A mass of 5 kg is attached to the end of a 1 m long uniform beam of mass 2 kg. The beam is fixed to a wall at one end and supported by a wire at an angle of 30° to the beam, 0.7 m

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from the wall. The beam is in equilibrium.



What is the tension *T* in the wire?

A. 17 N

B. 84 N

C. 140 N

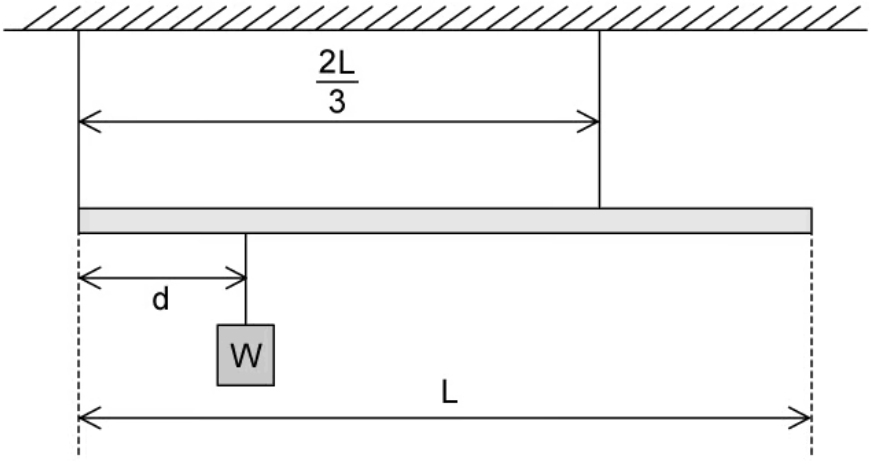
D. 170 N

(1 mark)

4 A horizontal beam of length L and weight $\frac{W}{2}$ is suspended from the ceiling by two ropes.

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One rope is attached at the left end of the beam, and the other is attached at a distance $\frac{2L}{3}$ from the left end. A load of weight W is suspended from the beam at a distance d from the left end, as shown. The beam is in equilibrium.



What value of d ensures that the tension is equal in both ropes?

A. $\frac{L}{8}$

B. $\frac{L}{4}$

C. $\frac{L}{3}$

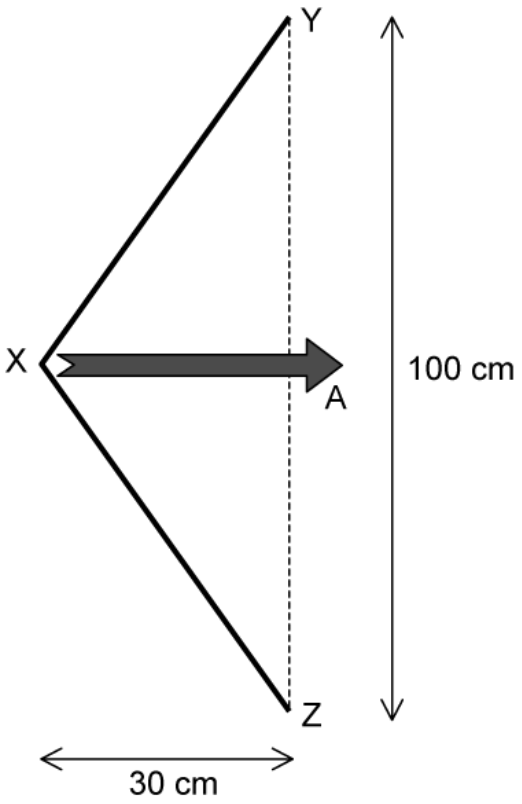
D. $\frac{L}{2}$

(1 mark)

5 The diagram shows a bow and arrow. The arrow is positioned at the centre X of the string of the bow stretched between points Y and Z. When the string is pulled back, the tension in the

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bowstring is 10 N.



What force will accelerate the arrow in the direction of point A when the string is released?

A. 5.2 N

B. 5.8 N

C. 8.6 N

D. 10.3 N

(1 mark)